

BUSINESS ANALYSIS REPORT

Instacart Market Basket Analysis

Key Findings and Business Recommendations

Based on exploratory analysis of ~32 million grocery order records

Executive Summary

THE PROBLEM

the business problem i set out to answer here is pretty simple to state but hard to actually answer without data: why do people keep coming back to certain products and not others, when are they actually placing orders, and can any of this be predicted well enough to plan staffing, inventory and retention around it. that's what this whole report is trying to work out, using close to 32 million order line items.

Short version of what i found: customers here are mostly creatures of habit, not explorers. 59% of everything ordered is a repeat purchase, and its concentrated almost entirely in stuff people run out of fast, milk, produce, water. Orders also aren't spread evenly through the week or the day, they pile up on Sunday and Monday, mostly between 9am and 4pm, and people even shop in a pattern within their own cart, routine stuff first, extras later.

None of this is just trivia, i think it actually tells the business exactly where to put its money and its people, and thats what the recommendations in each section below are built off of, not guesses.

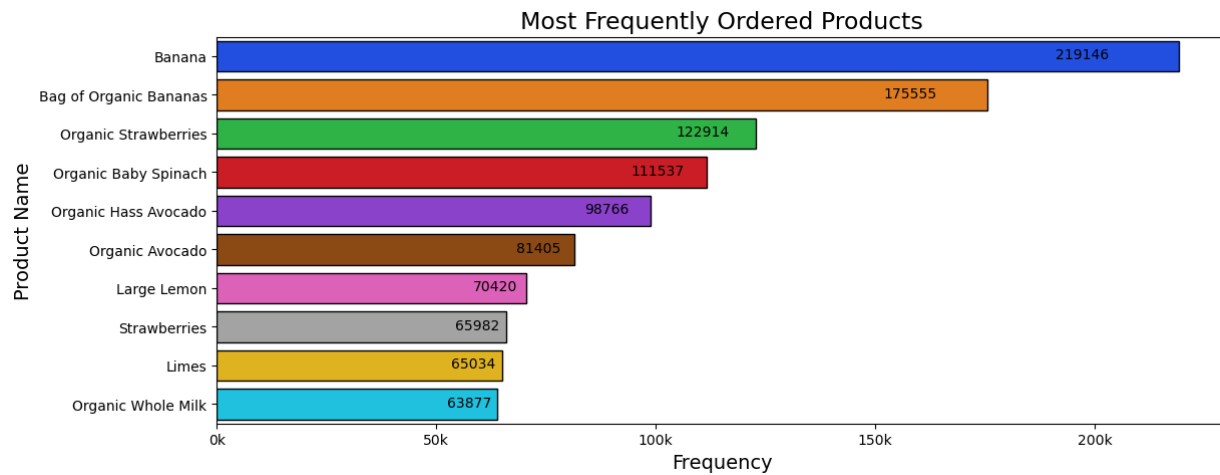
Metric	Value
Reorder share of all items	59% (vs. 41% first-time)
Top product / department	Banana (219,146 orders) / Produce (4.39M items)
Highest product reorder rate	Whole Organic Omega 3 Milk — 86%
Peak order day / window	Sunday (2.87M orders) / 9 AM – 4 PM
Average basket size / order cycle	10 items / ~13 days between orders
Reorder rate: 1st vs. 30th cart item	67.7% vs. 43.8%

1. Product and Category Performance

THE PROBLEM

the business doesn't really have hard proof of which products or categories deserve the shelf space, the ad budget or the inventory priority. before anyone spends more on any of that, i wanted actual evidence of whats driving volume and loyalty, not assumptions based on gut feel.

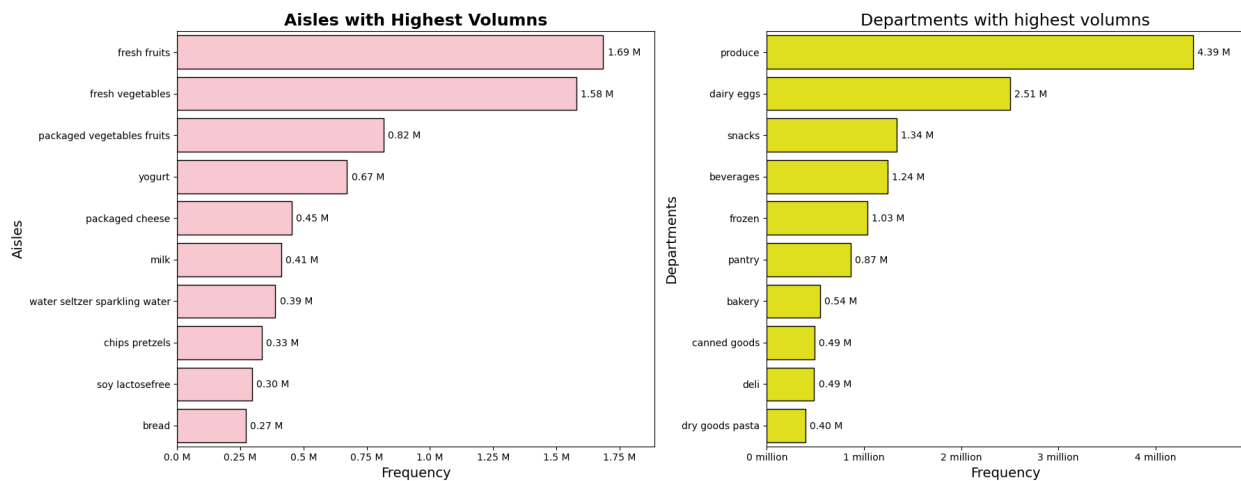
Q1. Which products are ordered most frequently?



MY TAKE

Bananas are, by a wide margin, the single most ordered item on the platform, 219,146 orders, with the bag of organic bananas right behind it. 9 of the top 10 products are produce, organic whole milk is the only outsider. **my recommendation:** protect produce availability above almost everything else, a stockout here costs the business more than a stockout almost anywhere else on the platform.

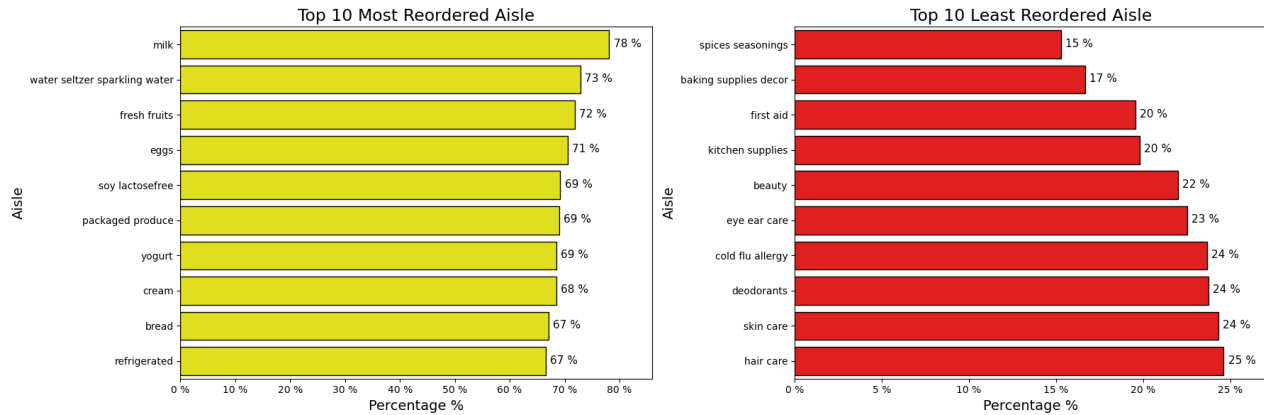
Q2. Which departments and aisles drive the most volume?



MY TAKE

Produce brings in 4.39 million items, almost double dairy eggs which comes in second at 2.51 million, and its mostly fresh fruits and fresh vegetables driving that number, not packaged goods. **so if i'm deciding where to put extra warehouse space** or push for better supplier rates, produce is where that conversation should start, not dairy or snacks.

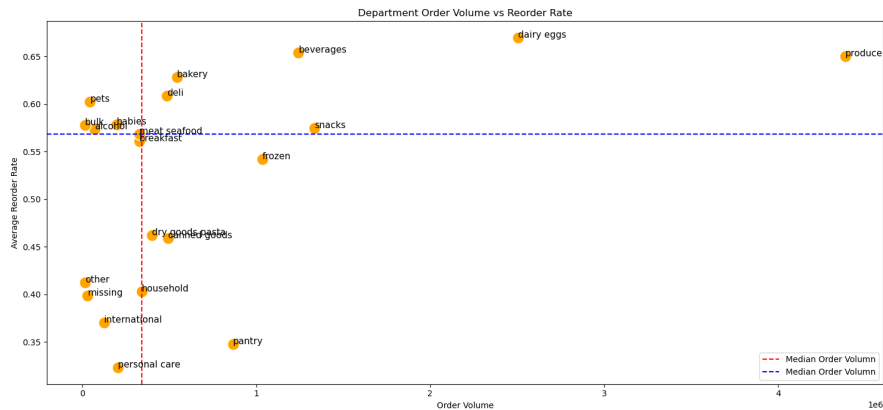
Q3. What is the reorder rate by aisle?



MY TAKE

Milk, water, fresh fruits and eggs top the reorder charts, all above 70%. spices and baking supplies barely reorder at all, 15-17%. reorder rate has basically nothing to do with how popular or expensive a category is, its about how fast people actually use the product up. **my recommendation:** push subscription or auto-reorder features hard on milk, water and produce, and don't bother pushing it on spices, it won't move the needle there.

Q4. Do high-volume departments show low reorder rates (novelty buys), or the reverse?



MY TAKE

I wanted to check if the top selling departments were just riding a wave of one-time curiosity buys, but that's not what's happening. Produce, dairy eggs and beverages are big AND loyal at the same time, one doesn't trade off against the other. **so the recommendation is simple,** keep doubling down on these three departments, they are not overvalued, the loyalty backs up the sales.

2. Customer Purchase Behavior

THE PROBLEM

delivery slots and retention campaigns shouldn't really be planned on a guess about how often people shop or how big their carts usually are, so i pulled the actual order frequency and basket size numbers to plan around instead of assuming.

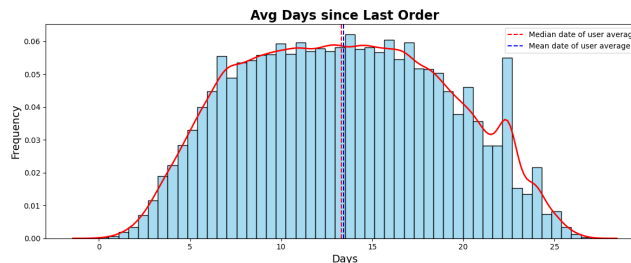
Q1. What is the distribution of orders per customer?



MY TAKE

Most customers on this platform are light users, the biggest group has only placed 4 to 6 orders total, not 50 or 100. There's a small loyal tail going up near 100, but that's very likely just a cutoff in how the dataset was recorded, not real behavior, so i wouldn't read too much into it. **my recommendation:** point retention spend at the huge base of light, occasional shoppers, not just the already-loyal few, that's where the actual growth is sitting.

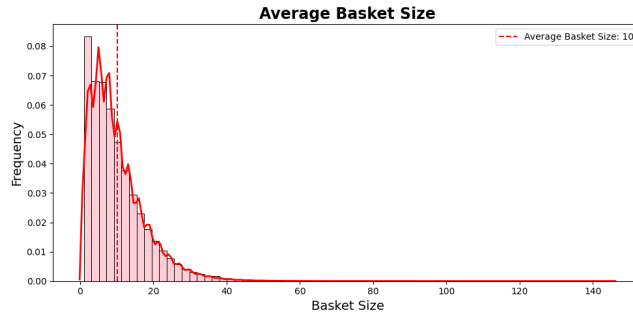
Q2. How many days pass between orders, on average?



MY TAKE

On average people reorder every 13 days, but it's not one clean number, there's actually two separate rhythms hiding in there, a weekly shopper group around the 7 day mark and a more monthly group around 22-23 days. **so a single one-size-fits-all reminder email probably isn't the right call**, i'd recommend at least two separate reorder-reminder schedules instead of one blanket one.

Q3. What is the average basket size?



MY TAKE

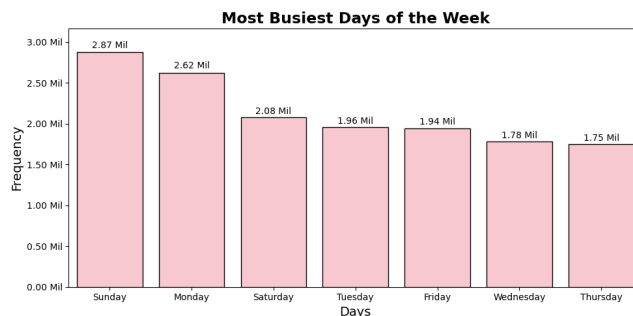
Average basket size comes out to 10 items but honestly that number is a bit misleading, most orders are actually small, just 1 to 5 items, its a handful of big bulk orders dragging the average up. **if delivery slots or packaging are being planned around a '10 item order,'** that's probably wrong for most orders, i'd plan fulfillment around the smaller, more common basket size instead.

3. Temporal and Seasonality Patterns

THE PROBLEM

i think staffing and delivery capacity is probably being set on gut feeling right now rather than actual demand, so i wanted to check when orders really spike, by day and by hour, and whether the two actually interact or not.

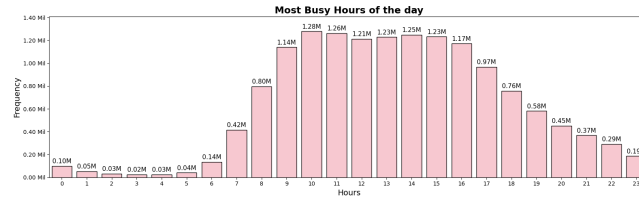
Q1. Which day of week is busiest?



MY TAKE

Sunday and Monday are clearly the two busiest days, 2.87 million and 2.62 million orders, and interestingly Saturday only comes in third, not second like you'd expect. Tuesday through Thursday is basically flat and quiet by comparison. **my recommendation:** schedule the bulk of delivery staff and restocking around Sunday and Monday specifically, not just 'the weekend' as a blanket assumption.

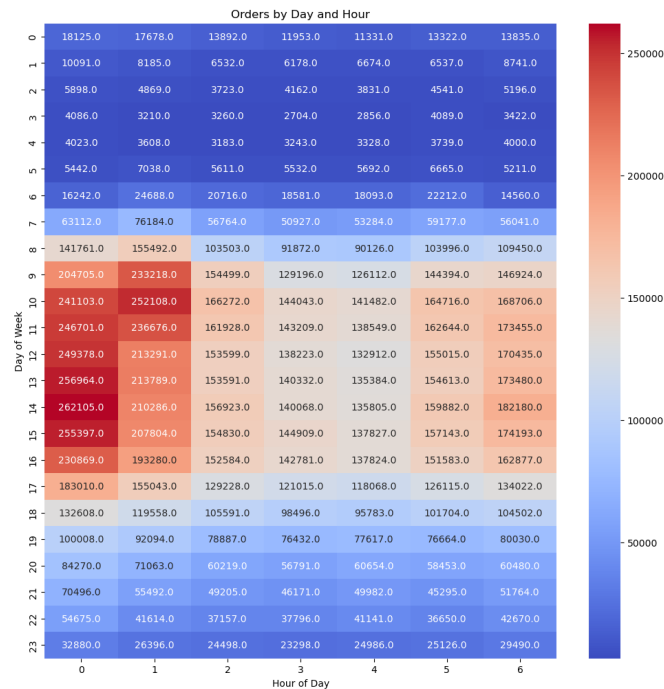
Q2. What hour of day sees peak activity?



MY TAKE

There isn't really one single peak hour, it's more of a long plateau from 10am to 4pm where order volume just stays high and steady. overnight is basically dead. **so staffing should reflect a wide, steady peak window**, not a short rush-hour spike, i'd keep strong coverage across the whole 10-4 stretch rather than concentrating everyone around lunchtime.

Q3. Do day and hour effects compound?



MY TAKE

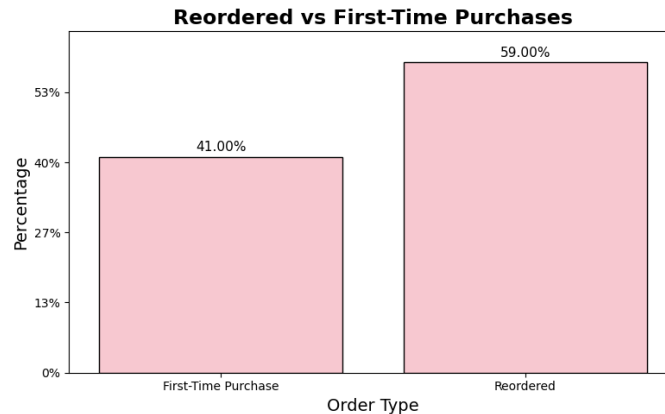
When i looked at day and hour together, the busiest slot by far is Sunday/Monday combined with the 10am-4pm window, over 260 thousand orders, higher than what you'd expect from either factor on its own. day and hour aren't really independent, they stack. **the recommendation is to plan staffing off this combined view**, not two separate day-only and hour-only schedules, the real peak is sharper than either view alone shows.

4. Reorder Behavior

THE PROBLEM

reorders are basically the backbone of repeat revenue on this platform, so i wanted to actually measure how big that behavior is, where it's concentrated, and whether something as simple as cart order can predict it.

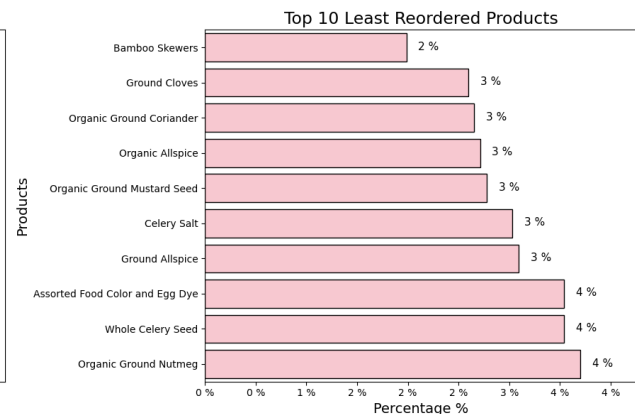
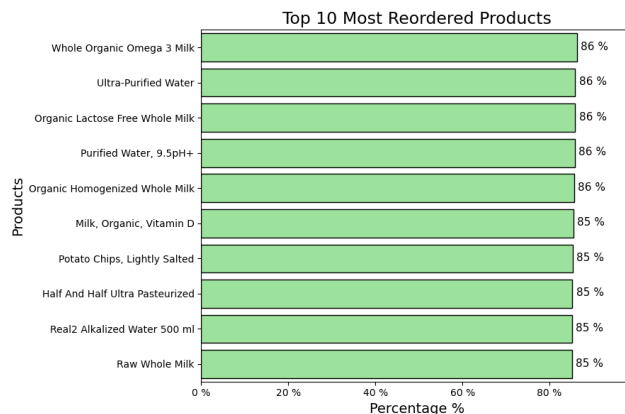
Q1. What share of items are reorders vs. first-time purchases?



MY TAKE

59% of everything ordered is a reorder, only 41% is a first time purchase. that's a big number honestly, it means the platform is mostly running on repeat habits already, not constant new discovery. **i think this is actually good news for the business**, retention has a strong base to build on, but it also means new customer acquisition needs its own separate push, it won't happen by accident off the back of reorders alone.

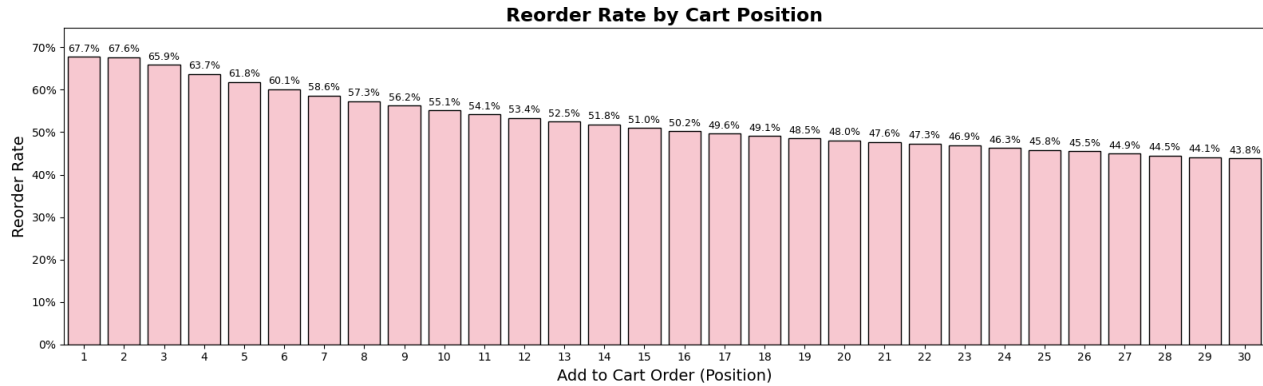
Q2. Which products have the highest reorder rate?



MY TAKE

Milk and water variants sit at the very top of the reorder list, 85-86%, while spice and specialty baking items like bamboo skewers or ground cloves barely reorder at all, 2-4%. again it's consumption speed driving this, not price or brand. **my recommendation:** build loyalty programs and subscribe options specifically around the milk/water tier, and don't waste marketing budget trying to get people to 'subscribe' to ground cloves, it just won't happen.

Q3. Does cart position predict reorder likelihood?



MY TAKE

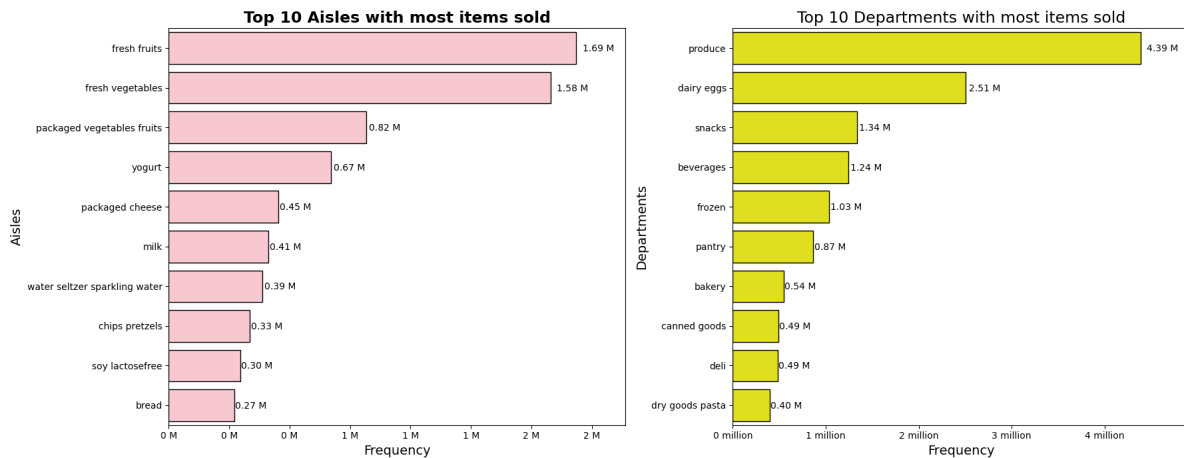
Reorder likelihood drops steadily the further down the cart you go, 67.7% for the very first item added, down to 43.8% by the 30th item. so people are clearly adding their usual stuff first and the extra or new stuff later in the trip. **i'd recommend using this pattern in the app itself**, show routine reorder suggestions right at the start of the session, and save new product recommendations for later in the cart, when people are already in more of an exploratory mindset.

5. Department and Aisle Segmentation

THE PROBLEM

category managers need a straight, data backed answer on where shelf space and marketing budget should actually go, not a guess based on which aisle just looks busiest on a given day.

Q1. Which aisles and departments sell the most items?



MY TAKE

This just confirms what showed up earlier, produce leads everything at 4.39 million items, with fresh fruits and vegetables doing most of the heavy lifting inside that department. **from a pure volume standpoint** there really isn't a case for treating any other department as equally important as produce right now.

Q2. What are the top 5 products per top department?



MY TAKE

Most departments are basically carried by one dominant product, Banana in Produce, Organic Whole Milk in Dairy Eggs, Olive Oil in Pantry. Snacks is the one exception, its top 5 products are much more evenly spread out, which tells me people are more experimental and less loyal to one brand in that category specifically. **so my recommendation is** to concentrate supply chain reliability on those single dominant products first, and treat snacks separately with more of a rotating, discovery-driven merchandising approach.

Overall, What I'd Actually Prioritize

if i had to rank everything above into an order the business should actually act on, this is roughly how i'd stack it:

- protect produce and dairy availability first, they carry both the volume and the loyalty, a stockout there costs more than anywhere else on the platform.
- build reorder / subscription pushes specifically around milk, water and produce, don't bother with low-reorder categories like spices, that effort is better spent elsewhere.
- staff and stock up hardest for Sunday-Monday, 10am-4pm, that's where demand genuinely spikes, not just 'the weekend' in general.
- run two separate reorder-reminder cadences, roughly weekly and roughly every three weeks, instead of one blanket schedule for everyone.
- redesign the in-app suggestion flow so routine items show up first in the cart experience, and discovery / new items show up later, matching how people actually shop.

A Couple Things This Report Doesn't Cover

- this is descriptive, i'm looking at what already happened, not predicting what happens next.
- the order-count cap at 100 per customer in the data is probably hiding just how loyal the top customers actually are, i wouldn't treat that number as a real ceiling.
- i didn't get into which products get bought together in the same basket (market basket / association rules), that's a natural next step from here.
- a proper cohort/retention curve, how many first-time customers come back for a 2nd, 5th, 10th order, would sharpen the retention recommendations even further.